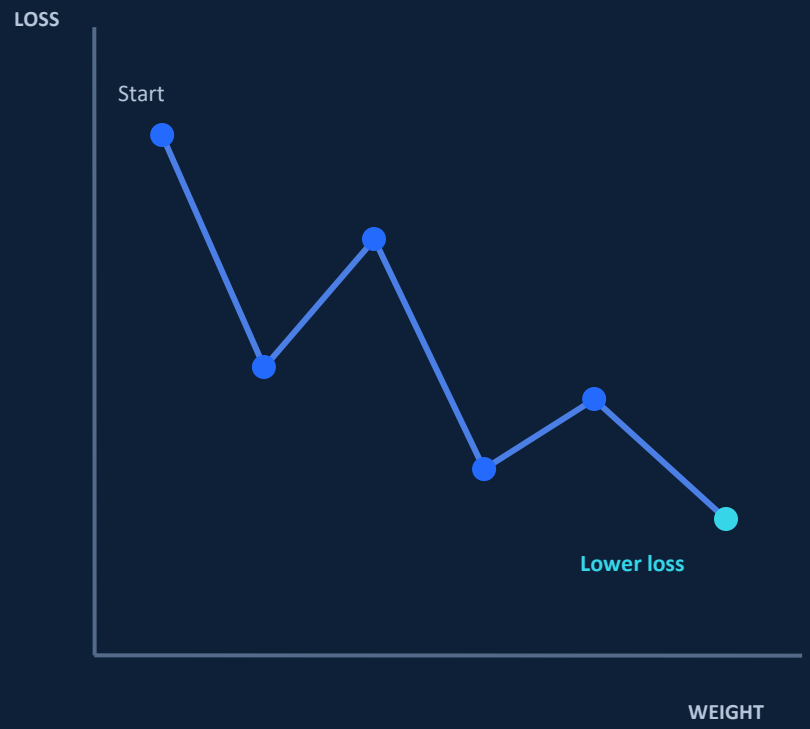
STOCHASTIC GRADIENT DESCENT REGRESSOR

Understanding SGD optimization and regression modelling with Scikit-learn.

> SGD updates model weights from sample-level gradients, moving toward lower prediction loss.

LOSS LANDSCAPE



$$w_{t+1} = w_t - \eta \cdot \nabla L(w_t)$$

What is SGDRegressor?

A linear regression estimator trained with stochastic gradient descent.

- **Predicts continuous values**
  Examples include salary, house prices, electricity use and sales.

- **Learns linear coefficients**
  Repeated updates estimate feature weights and an intercept.

$$\hat{y} = w_0 + w_1x_1 + w_2x_2 + \ldots + w_px_p$$

MODEL LEARNING



SGD	Optimization algorithm
SGDRegressor	Regression estimator that uses SGD

Loss functions and regularization

Loss measures prediction error and supplies the signal for parameter updates.

ONE-SAMPLE SQUARED ERROR

$$(y - \hat{y})^2$$

Loss choices

They change how prediction errors contribute to the objective.

Loss	Behaviour	Typical use
squared_error	Squares all residuals	Standard regression
huber	Quadratic near zero, linear for large errors	More robust to outliers
epsilon_insensitive	Ignores errors within ±ε	Tolerance-based regression
squared_epsilon_insensitive	Squares errors outside ±ε	Larger errors need stronger penalty

Loss function measures prediction error

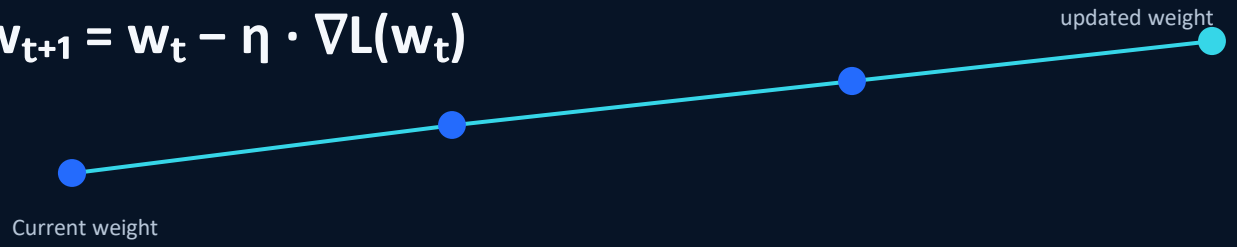
Regularization penalty adds a complexity cost to control coefficient size

Gradients, epochs and learning rate

- **Gradient**
  Direction and rate at which loss changes as a parameter changes.
- **Learning rate (η)**
  Sets the size of each parameter update.
- **Epoch / max_iter**
  One full pass over the data; max_iter sets the epoch limit.
- **tol and eta0**
  tol controls convergence stopping. eta0 initializes supported schedules.

PARAMETER UPDATE

$$w_{t+1} = w_t - \eta \cdot \nabla L(w_t)$$



Update granularity

Batch	Stochastic	Mini-batch
All examples per update	One example per update	Small subset per update

Scikit-learn SGDRegressor updates sample by sample in fit(). It has no batch_size parameter.

Parameters and training workflow

Set the objective, regularization strength and optimization limits deliberately.

CORE PARAMETERS

- **loss** — Prediction error objective
- **penalty** — l1, l2, elasticnet or none
- **alpha** — Regularization strength
- **l1_ratio** — L1 share for elastic net
- **learning_rate / eta0** — Schedule and initial step size
- **max_iter / tol** — Epoch limit and stopping rule
- **random_state** — Repeatable shuffling

PYTHON / SCIKIT-LEARN

```
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import SGDRegressor

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

model = SGDRegressor(
   loss='squared_error', penalty='l2',
   alpha=0.0001, max_iter=1000,
   tol=1e-3, random_state=42
)
model.fit(X_train_scaled, y_train)
y_pred = model.predict(X_test_scaled)
```

REGULARIZATION

L1 can set coefficients to zero. L2 shrinks coefficients. Elastic Net combines both.

WORKFLOW

Feature scaling — **Model creation** — **Training** — **Prediction** — **Evaluation**

KEY TAKEAWAYS

UNDERSTANDING SGDREGRESSOR

01 **SGD**

The optimization algorithm

02 **SGDRegressor**

The regression estimator

03 **Loss + gradient**

Measure error and guide updates

04 **Learning rate**

Controls update size

05 **Regularization**

Helps control model complexity

06 **Feature scaling**

Supports stable learning

Understand the optimization → Train the model → Evaluate the predictions.